African Institute for Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 11: Confidence Intervals

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1. Useful results

2. Confidence Intervals

Combinations of r.v.'s

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Chi-square distribution

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Definition

A r.v. X has a ***chi-square*** *distribution with n degrees of freedom (d.o.f.) iff its p.d.f has the form:*

$$f_X(x) = \frac{2^{-n/2}}{\Gamma(n/2)}\, x^{n/2-1}\, e^{-x/2},\ x \in \mathbb{R}^+ \ , n \in \mathbb{N}^*.$$

Notation: $X \sim \chi^2(n)$. The c.d.f. is denoted: $\chi^2_n(x) = P(X \leqslant x)$.

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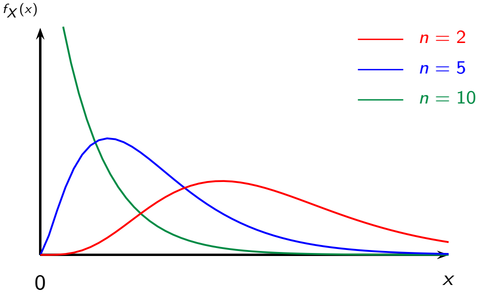
The p.d.f. of a **Chi-square** r.v. with n d.o.f. is the p.d.f of a **gamma distribution** with parameters $(n/2, 1/2)$.

Chi-square distribution

P.d.f of a Chi-2 r.v for different d.o.f.



Combinations of r.v.'s

- **Student's t-distribution**
  - Let $Z \sim \mathcal{N}(0,1)$ and $Y \sim \chi^2(n)$ be two indepedent r.v.'s, then:

$$T = \frac{Z}{\sqrt{Y/n}}$$

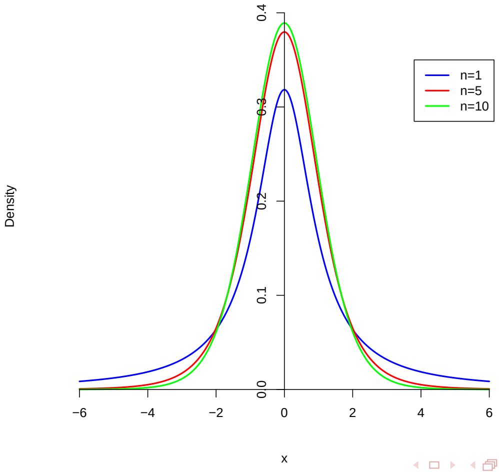
    follows a **Student's t-distribution** with n degrees of freedom.
  - Notation: $Z \sim \mathscr{T}(n)$.
  - The p.d.f. of a **Student's *t*-distribution** is:

$$f_T(t) = \frac{\Gamma[(n+1)/2]}{\sqrt{n\,\pi}\,\Gamma(n/2)} \cdot \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}, \quad t \in \mathbb{R}^+.$$

  - The c.d.f. is denoted: $t_n(x)$.

The Student's t-distribution

P.d.f. of the Student's distribution for $n = 2, 5, 10$

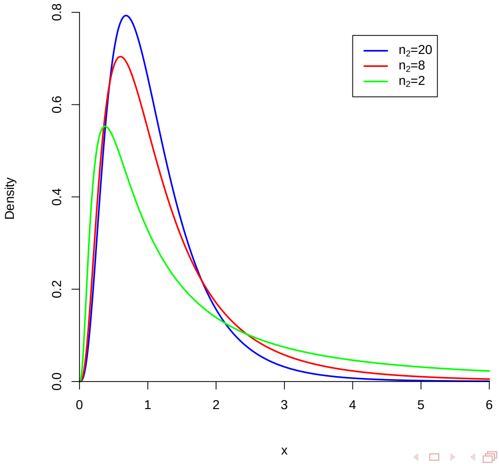


Combinations of r.v.'s

- **Fisher distribution**
  - Let $Y_1 \sim \chi^2(n_1)$ and $Y_2 \sim \chi^2(n_2)$, be two independent r.v.'s, then
$$Y = \frac{Y_1/n_1}{Y_2/n_2}$$
follows a **Fisher distribution** with (n_1, n_2) degrees of freedom.
  - Notation: $Y \sim \mathscr{F}(n_1, n_2)$.
  - The p.d.f. of a **Fisher distribution** with (n_1, n_2) degrees of freedom is:
$$f_Y(y) = \frac{\Gamma[(n_1+n_2)/2]}{\Gamma(n_1/2)\,\Gamma(n_2/2)} \cdot \left(\frac{n_1}{n_2}\right)^{n_1/2} \cdot \frac{y^{(n_1-2)/2}}{(1+n_1\ y/n_2)^{(n_1+n_2)/2}}.$$

The Fisher distribution

P.d.f. of the Fisher distribution for $n_1 = 8$ *and* $n_2 = 20, 8, 2$



Joint moments generating function

Definition

Let $X_1, X_2, \ldots, X_n$ be n r.v.'s. The joint moments generating function of $(X_1, \ldots, X_n)$ is:

$$M_{X_1,\ldots,X_n}(t_1, \ldots, t_n) = E\left(e^{t_1 X_1 + \ldots + t_n X_n}\right),$$

where $t_1, \ldots, t_n$ are real numbers.

Proposition

Let X_1, X_2 be two r.v.'s. X_1 and X_2 are independent iff the joint moments generating function of X_1, X_2 is the production of m.g.f. of X_1 and X_2.

$$M_{X_1,X_2}(t_1, t_2) = M_{X_1}(t_1)\ M_{X_2}(t_2),\ \forall t_1, t_2.$$

Confidence Intervals

Definition

A $(1-\alpha)$-confidence interval (C.I.) for a parameter θ is an interval in $\Theta \subseteq \mathbb{R}$ that has a probability $1-\alpha$ to contain the true value of θ.

- Constructing a $(1-\alpha)$-confidence interval is finding an interval (a, b) such that $P\big((a,b) \ni \theta\big) = 1-\alpha$ that is

$$P(a < \theta \leqslant b) = 1 - \alpha.$$

- The methods will rely on the properties of the estimators.

First strategy: pivotal function

Definition

A pivotal function is a statistic depending on the parameter of interest such that its probability distribution does not depend on the unknown parameters.

Example:
Suppose that a sample $X_1, \cdots, X_n$ of n independent r.v.'s from an exponential distribution with parameter θ.

To build a C.I. for θ, we consider the pivotal function: $W = \dfrac{2n\overline{X}_n}{\theta}$.

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Possible bounds: $a = \dfrac{2n\overline{X}_n}{\chi_{2n}^{2^{-1}}(1-\alpha/2)}$ and $b = \dfrac{2n\overline{X}_n}{\chi_{2n}^{2^{-1}}(\alpha/2)}$.

First strategy: pivotal function

Computation example with R

```
> theta<-38.4
> n<-50
> x<-rexp(n,1/theta)
> alpha<-0.05
> alpha1<-alpha2<-0.025
> a<-2*n*mean(x)/qchisq(1-alpha1,2*n);a
[1] 26.94047
> b<-2*n*mean(x)/qchisq(alpha2,2*n);b
[1] 47.02707
```

➥ The interval [26.94 ; 46.93] has a probability 0.95 to contain the true value of the parameter *theta*.

First strategy: pivotal function

Example: C.I for the parameters of a normal distribution

Theorem

Let $X_1, \cdots, X_n$ be n i.i.d. r.v.'s from $\mathcal{N}(\mu, \sigma^2)$.

(*i*) $\overline{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$.

(*ii*) $(n-1)\widehat{\sigma}_n^2/\sigma^2 \sim \chi^2_{n-1}$.

(*iii*) $\overline{X}_n$ *and* $(n-1)\widehat{\sigma}_n^2/\sigma^2$ *are independent r.v.'s.*